

EU carbon border, Indian steel, 2026 to 2034

Verify first, route second, price at home third

What CBAM costs India's five largest steel exporters, and the five moves that take a €14.7bn bill to €2.8bn. Board and Ministry readout, 9 November 2026.



€14.7bn

bill 2026 to 2034 on EU default values

66%

of it is a charge on missing data

€2.8bn

left after five moves

Built from the cbam-india repository and live site, **cbam-india.vercel.app**. Every number is recomputed in the deck ledger and ties to the repo; 2026 context added from sources dated to 8 Nov 2026.

Independent analysis. Producer intensities and EU volumes are analyst estimates (grade D): replace with verified installation data before relying on a single number.

Free allocation left

2026	2027	2028	2029	2030	2031	2032	2033	2034
97.5%	95%	90%	77.5%	51.5%	39%	26.5%	14%	0%

Two thirds of a €14.7bn CBAM bill is a charge on missing data; five moves cut it to €2.8bn by 2034

Five exporters, 2026 to 2034, EU importer's legal cost at 100% pass through unless stated. The bill is small now and steep after 2030.

1 Bill 2026 to 2034

€bn, five exporters

€14.7bn on EU default values

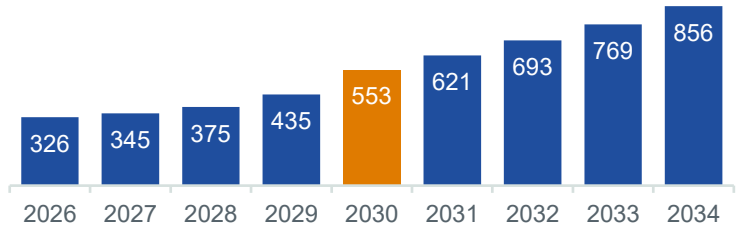
€4.97b on verified data



66% of the default bill disappears with verified emissions data; the rest is real carbon above the EU benchmark.

2 Annual bill on verified data

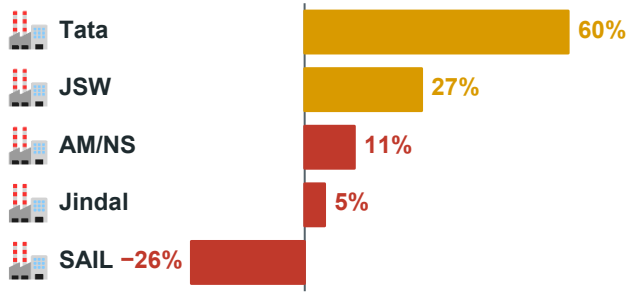
€m a year



2.6x €326m in 2026 to €856m in 2034; the 2030 step alone adds €118m.

3 EU margin kept in 2030

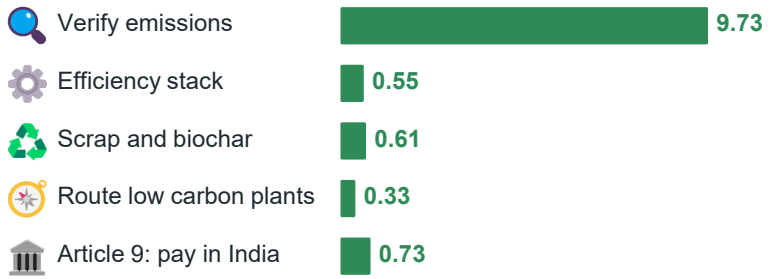
% of EBITDA



At 50% pass through SAIL's EU tonnes lose money by 2030; Jindal keeps 5%, AM/NS 11%.

4 Five moves, removed 2026 to 2034

€bn



Residual €2.75bn paid to the EU, 19% of the default bill.

5 What changed since the repo (31 Oct)

dated 2026

1.94 Mt	duty free EU quota from 1 Jul (was 2.79); 50% duty above
−19.9%	steel to the EU, Apr to Sep FY27, 1.53 Mt
Nov	EU act on deducting carbon paid abroad due
Nov	CCTS certificates start trading
0	CBAM exemptions in the India EU FTA

6 Decisions this quarter

owner

1	**Verify 100% of EU tonnes by 2027** worth €739m a year to EU buyers	Company
2	**Index top 10 EU contracts to the certificate price** share the cost by clause, not by discount	Company
3	**Plant CBAM cost in S&OP and quota slots** routing worth €10m a year to JSW alone	Company
4	**Explicit €20/t export price, Art. 9 evidence** keeps €974m in India vs €49m	Government

Source: Reg. (EU) 2023/956; Directive 2003/87/EC Art. 10a; Impl. Regs (EU) 2025/2620 (benchmarks), 2025/2621 and 2026/1740 (defaults, mark ups); DG TAXUD certificate prices; cbam-india model (31 Oct 2026), ported and tied out in this deck's ledger; quota: LSE Grantham on Impl. Reg. (EU) 2026/1457; exports: Commerce Ministry via Moneycontrol; Art. 9 and CCTS: Carbon Pulse, CERC, BEE.

All term moved

Verdict Verification is the first and largest lever; it also decides which tonnes can still use the smaller EU quota.

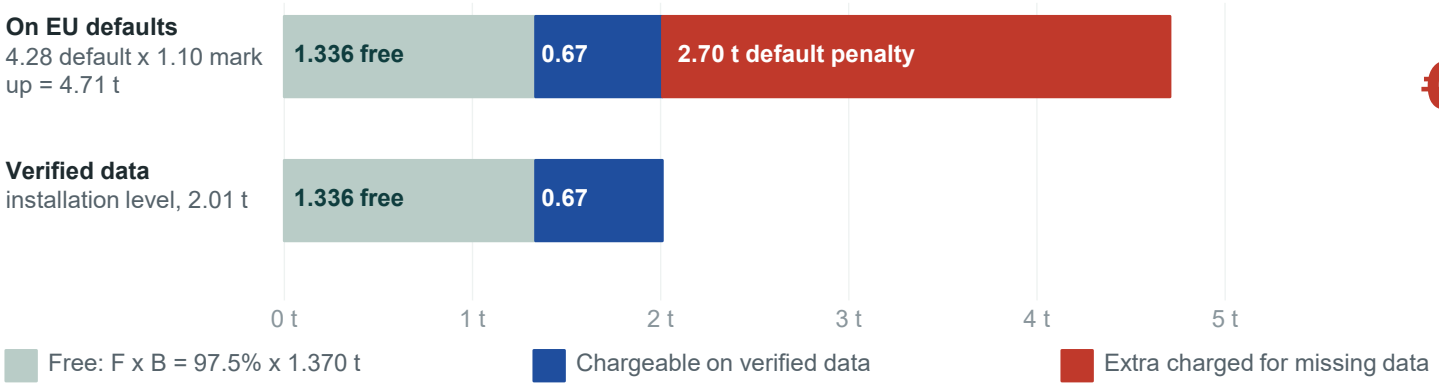
Next How one tonne is charged

One tonne of Tata hot rolled coil pays €278 on EU defaults and €56 verified: the gap is data, not carbon

Certificates per tonne = max(0, E − F×B); cost = certificates × P − any carbon price paid in India (Art. 9). The 2.5% cut in 2026 applies to the benchmark, not the bill.

1 One tonne of Tata hot rolled coil (CN 7208), 2026

tCO2 per t of steel, bars to scale



3.37 t x €82.32
€278

0.67 t x €82.32
€56

2 What data is worth

€, 2026

€222/t
charged on one Tata tonne for missing verified data

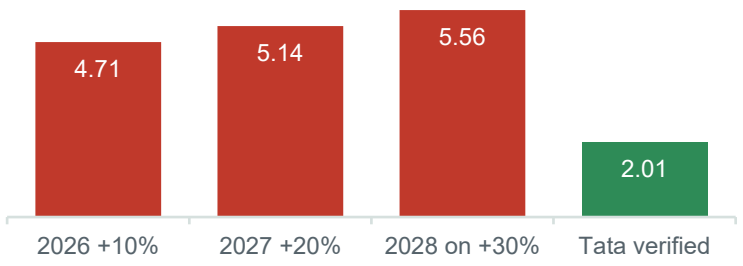
€200m
a year, Tata EU tonnes

€739m
a year, five exporters

Value to the EU importer of verified data at today's price; the repo puts verification cost at tens of thousands of € per installation.

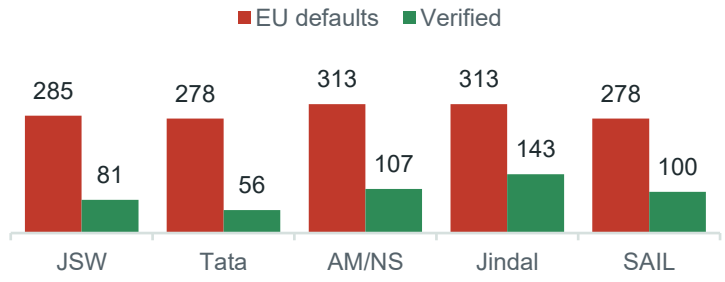
3 Default mark up steps up

t applied per t



4 Defaults vs verified, five mills

€ per t, 2026



5 Who pays, and when

Tata, € per t



Exporter share at 50% pass through: **€28 verified, €139 on defaults**. Certificate sales open Feb 2027; first declaration for 2026 imports due 30 Sep 2027.

Source: Reg. (EU) 2023/956; Directive 2003/87/EC Art. 10a; Impl. Regs (EU) 2025/2620 (benchmarks), 2025/2621 and 2026/1740 (defaults, mark ups); DG TAXUD certificate prices; Tata intensity 2.01 t is a CCTS plant baseline x 0.85 direct share (grade D); cbam-india model (31 Oct 2026), ported and tied out in this deck's ledger.

Free allocation left

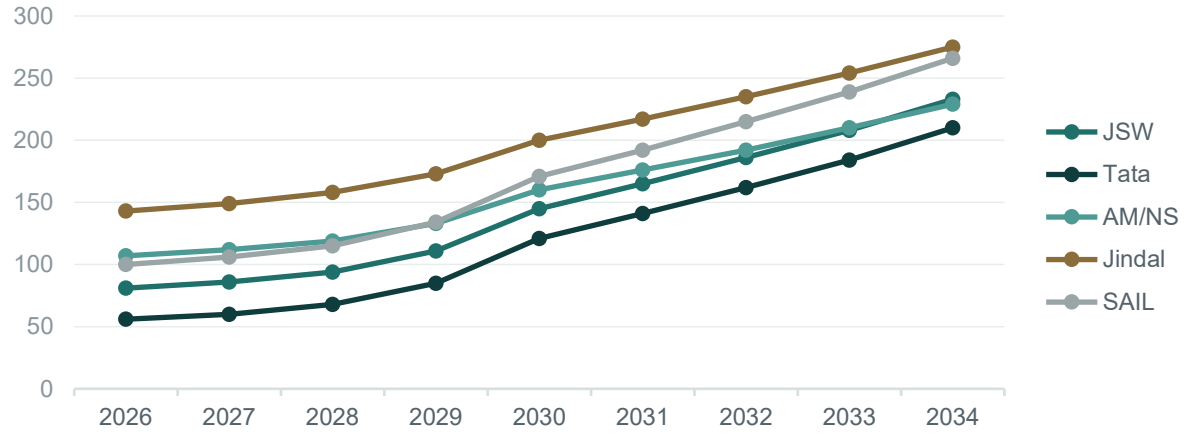
2026	2027	2028	2029	2030	2031	2032	2033	2034
97.5%	95%	90%	77.5%	51.5%	39%	26.5%	14%	0%

The bill is back loaded: free allocation falls from 77.5% to 51.5% in 2030, and verified steel pays €210 to €275 a tonne by 2034

Verified data, 100% pass through; certificate €82.32 in 2026, then +3% a year (a scenario, not a forecast).

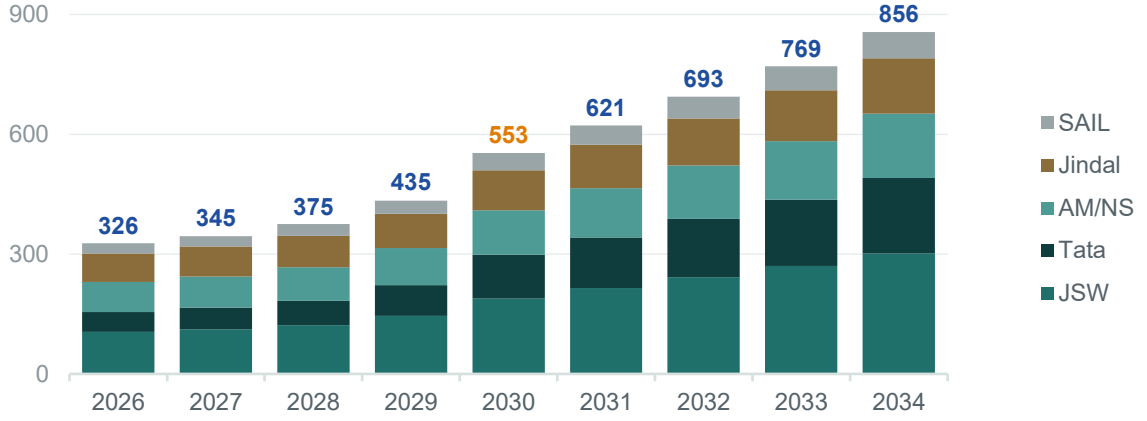
1 Verified CBAM cost per tonne

€ per t, 2026 to 2034



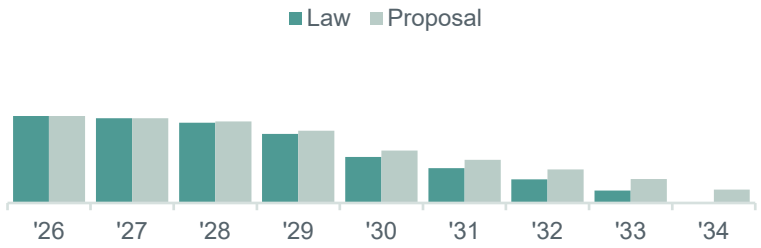
2 Annual bill on verified data, by exporter

€m a year



3 Free allocation path

% of benchmark free



4 The 2030 step

2029 to 2030

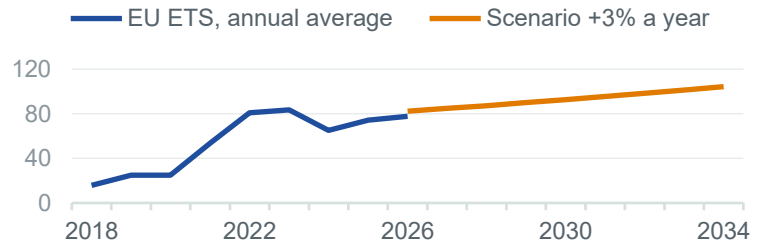
−26 pts
free allocation, 77.5% to 51.5%

+42%
Tata per tonne, €85 to €121

+€118m
five exporter bill, to €553m

5 Carbon price: history and scenario

€ per tCO2



Source: Reg. (EU) 2023/956; Directive 2003/87/EC Art. 10a; Impl. Regs (EU) 2025/2620 (benchmarks), 2025/2621 and 2026/1740 (defaults, mark ups); DG TAXUD certificate prices; ETS history: repo roadmap.ts (annual averages); proposal COM(2026) 616 is not law; cbam-india model (31 Oct 2026), ported and tied out in this deck's ledger.

F x B
term moved

Verdict Every move that cuts tonnes must land before 2030: after it, each tonne above the benchmark costs about twice as much as in 2026.

Next Who is exposed

Free allocation left

2026	2027	2028	2029	2030	2031	2032	2033	2034
97.5%	95%	90%	77.5%	51.5%	39%	26.5%	14%	0%

By 2030 AM/NS and Jindal keep 11% and 5% of their EU margin, and SAIL's EU tonnes lose money

Five tests with stated thresholds, so the call can be reproduced. DRI routes get the 0.481 DRI-EAF benchmark, so a DRI tonne gets less free allocation than a blast furnace tonne at the same emissions.

1 Executive scorecard

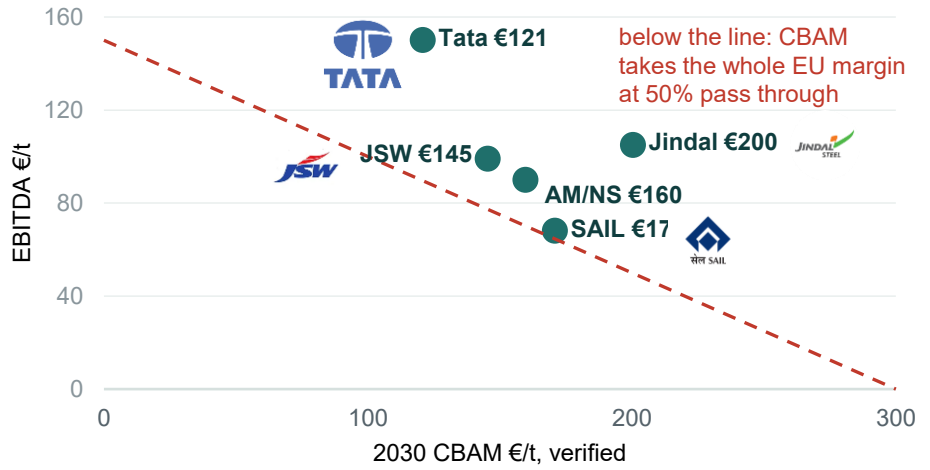
red, amber, green by stated rule

	Cost 2030 red >€150 amber >€100	Margin kept 2030 red <25% amber <60%	Default penalty 2026 red >€200 amber >€100	DRI share red >40% amber >15%	2030 vs 2026 red >3x amber >2x
JSW Steel	€145	27%	€204/t	10%	1.8x
Tata Steel	€121	60%	€222/t	0%	2.2x
AM/NS India	€160	11%	€206/t	50%	1.5x
Jindal Steel	€200	5%	€170/t	50%	1.4x
SAIL	€171	−26%	€178/t	0%	1.7x

Red flags: AM/NS 4, Jindal 3, SAIL 2, JSW 1, Tata 1. Margin test at the base 50% pass through.

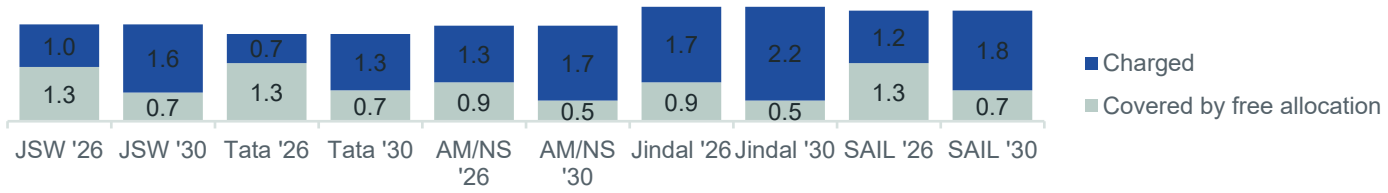
2 2030 cost against EU margin

€ per t



3 Free allocation covers less each year

tCO2 per t: covered and charged



4 EU margin per tonne left in 2030

€ per t, 50% pass through

JSW	Tata	AM/NS	Jindal	SAIL
€26	€90	€10	€5	−€17
EBITDA €99	EBITDA €150	EBITDA €90	EBITDA €105	EBITDA €68

Source: Thresholds and inputs: repo exhibits.ts scorecard; EBITDA per t from FY26 results (JSW, Tata, JSPL, SAIL grade C; AM/NS grade D); intensities and EU volumes grade D; cbam-india model (31 Oct 2026), ported and tied out in this deck's ledger.

E and B term moved

Verdict Exposure is about route and margin, not size: the two DRI heavy mills and SAIL need verification and routing first.

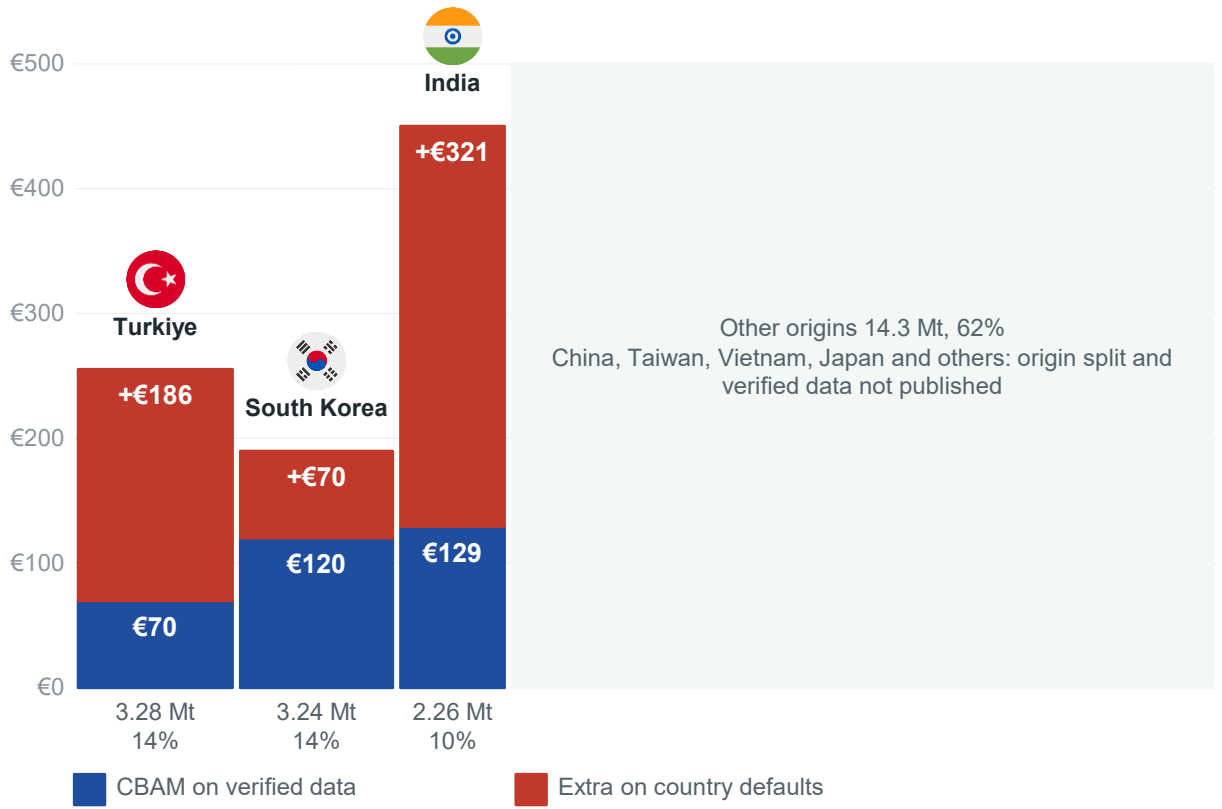
Next Against other origins

Verified, India pays €129/t in 2030 against Korea's €120; on defaults it pays €321/t more and loses to EU mills above €19/tCO2

Width is each origin's share of 23.1 Mt of EU flat steel imports in 2025; height is the CBAM cost per tonne in 2030. Delivered costs are illustrative 2026 FOB and freight levels (grade D).

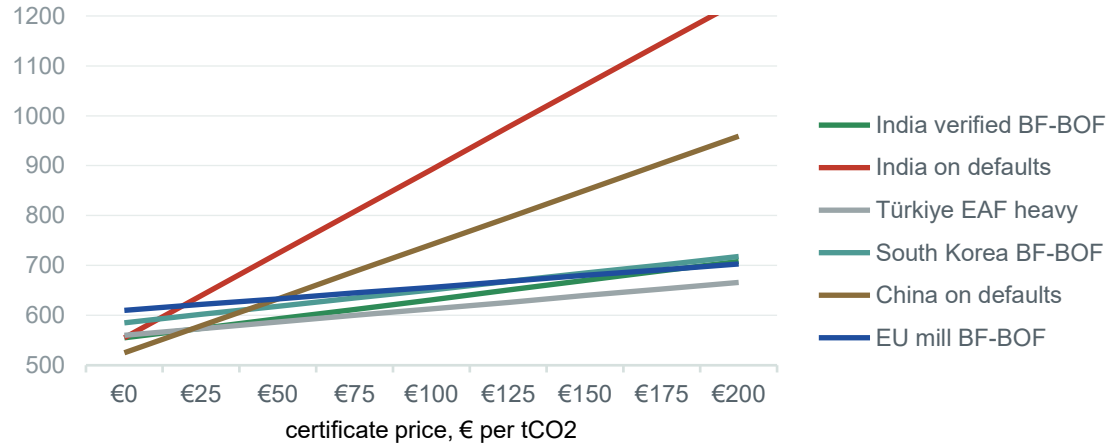
1 EU flat steel imports by origin and CBAM cost, 2030

x: Mt 2025; y: € per t



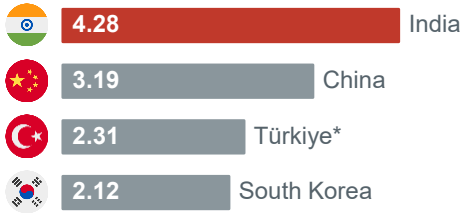
2 Delivered cost into the EU vs carbon price

€ per t of HR coil, 2026 rules



3 Country default values

tCO2 per t



4 Where India loses

€ per tCO2

€19
on defaults, vs an EU mill

€183
verified, vs an EU mill

Today's certificate: €82.32

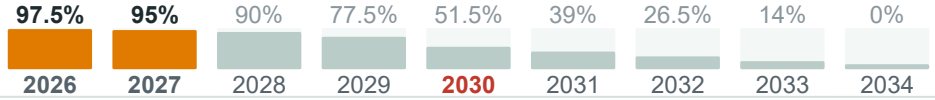
Source: EUROFER European Steel in Figures 2026 via cbam.clarion.engineer (imports, grade C); country defaults: Impl. Reg. (EU) 2026/1740, cbamguide.com (Turkiye bars and rods default as proxy); FOB, freight, peer intensities grade D; cbam-india model (31 Oct 2026), ported and tied out in this deck's ledger.

E term moved

Verdict With verified data India competes with Korea; on defaults it is priced out of the EU at any carbon price above €19.

Next What changed in 2026

Free allocation left



Since July a 1.94 Mt duty free quota rations Indian steel; the margin left after CBAM decides which tonnes should use it

The repo holds EU volumes flat to 2034. Since then the quota, falling shipments and the Article 9 and CCTS calendar have moved.

1 India's EU steel quota

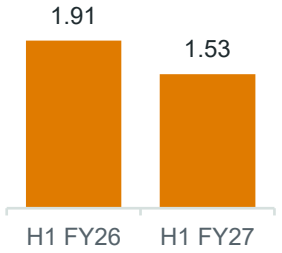
Mt a year



−30% country quota; 50% duty above it, on top of CBAM

2 Indian steel shipped to the EU

Mt, Apr to Sep



Jul FY27	**0.32 Mt**	**−2.3%**
Aug FY27	**0.39 Mt**	**flat**
Sep FY27	**0.20 Mt**	**−50.4%**

−19.9% year on year; to the Americas +262%

3 The 2026 to 2027 calendar

law, draft, expected

27 Jan 26	FTA concluded: no CBAM exemption
13 May 26	Draft act on carbon paid abroad
1 Jul 26	Safeguard quota 2.79 to 1.94 Mt
Nov 26	Art. 9 act and default carbon prices due
Nov 26	CCTS certificates trade on exchanges
1 Feb 27	CBAM certificate sales open
30 Sep 27	First declaration, 2026 imports

4 If the quota binds: EU margin left per tonne after CBAM, 2030

our assessment, € per t



Five exporters ship 3.65 Mt in the repo baseline, more than the country quota. The quota is first come at the border, not split by mill: this ranks which tonnes stay worth shipping. Out of quota, a 50% duty on a €600 coil (€300) beats every margin here.

5 What it means

for the bill

- Per tonne cost now decides access: Tata keeps €90/t in 2030, SAIL loses €17/t.
- At the H1 run rate (−19.9%) the verified bill falls to €3.98bn, but the cost per tonne does not move.
- Art. 9 rules and CCTS trading land this winter: the state's move is live now.

Source: LSE Grantham Institute on Implementing Reg. (EU) 2026/1457 (from 1 Jul 2026); govt official, Sep 2026, on residual pools; exports: Commerce Ministry data via Moneycontrol, Apr to Sep FY27; Art. 9: Commission draft (13 May 2026), Carbon Pulse (23 Oct 2026); CCTS: CERC CCC Regulations 2026, BEE, ICAP; FTA: Commerce Ministry (27 Jan 2026). Module 4 is our assessment.

Volume
term moved

Verdict Quota tonnes are now scarce: give them to verified, low carbon plants and stop counting on volume to absorb CBAM.

Next Which levers pay

Free allocation left

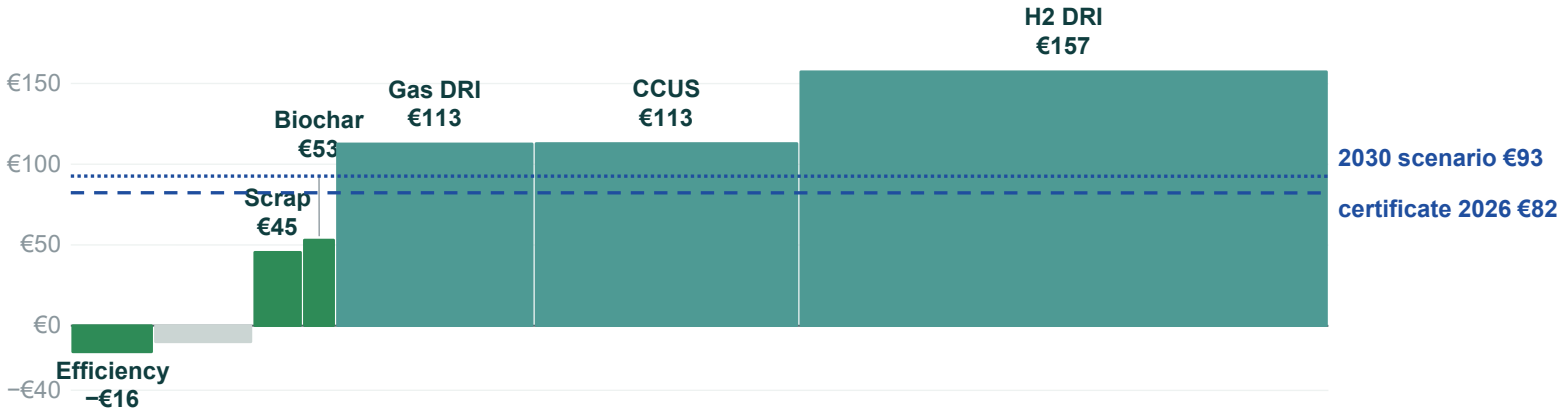
2026	2027	2028	2029	2030	2031	2032	2033	2034
97.5%	95%	90%	77.5%	51.5%	39%	26.5%	14%	0%

Efficiency pays on its own; scrap and biochar cost €45 to 53 a tonne of CO2 against an €82 certificate; a renewable PPA cuts no steel CBAM

Abatement cost curve for JSW's EU bound tonnes: width is tCO2 a year abated, height is the cost per tCO2 before any CBAM saving (10% discount rate, lever life).

1 Abatement cost curve, JSW EU tonnes

x: MtCO2 a year; y: € per tCO2



4.94 MtCO2 a year on JSW's 1.3 Mt of EU tonnes if every lever ran. Grey: renewable PPA at −€10/tCO2 cuts real emissions, not the steel CBAM bill.

2 Lever economics, JSW

NPV € per t of capacity

Lever	Cut t/t	Group EU share	Dedicated EU line	TRL
Efficiency stack	0.20	€34	€107	9
RE PPA	0.00	€27	€27	9
Scrap in BOF	0.15	−€40	€2	9
Biochar in BF	0.10	−€31	−€3	7
Gas DRI-EAF	0.60	−€600	−€320	9
CCUS	0.80	−€753	−€414	6
Green H2 DRI	1.60	−€2,253	−€1,557	6

Dedicated EU line: every tonne of the lever earns the CBAM saving.

3 A PPA saves no steel CBAM

Annex II

€0

steel CBAM saved

Steel and aluminium pay on direct emissions only. A PPA cuts 0.3 t of Scope 2: it counts for CCTS and buyers, not CBAM. Cement is the exception.

4 Aluminium: a tail risk

Vedanta, € per t, 2026



Captive coal power, about 14.5 MWh per tonne, sits outside the charge today.

5 Hydrogen DRI is an option

per t of capacity

€157 per tCO2 abated

−€1,557 NPV on a dedicated EU line

Capex €650/t, extra opex €180/t, TRL 6. Reserve site and power now; go or no go in 2030 on certificate price, H2 cost and a green premium.

Source: Lever ranges: IEA Iron and Steel Technology Roadmap; Ministry of Steel, Greening the Steel Sector (2024); TERI, CSTEP (grade D); Annex II of Reg. (EU) 2023/956 (direct emissions only for iron, steel and aluminium); cbam-india model (31 Oct 2026), ported and tied out in this deck's ledger.

E term moved

Verdict Spend first where a tonne of CO2 costs less than the certificate; treat gas DRI, CCUS and hydrogen as options priced on a premium, not on CBAM.

Next Value at stake

Free allocation left

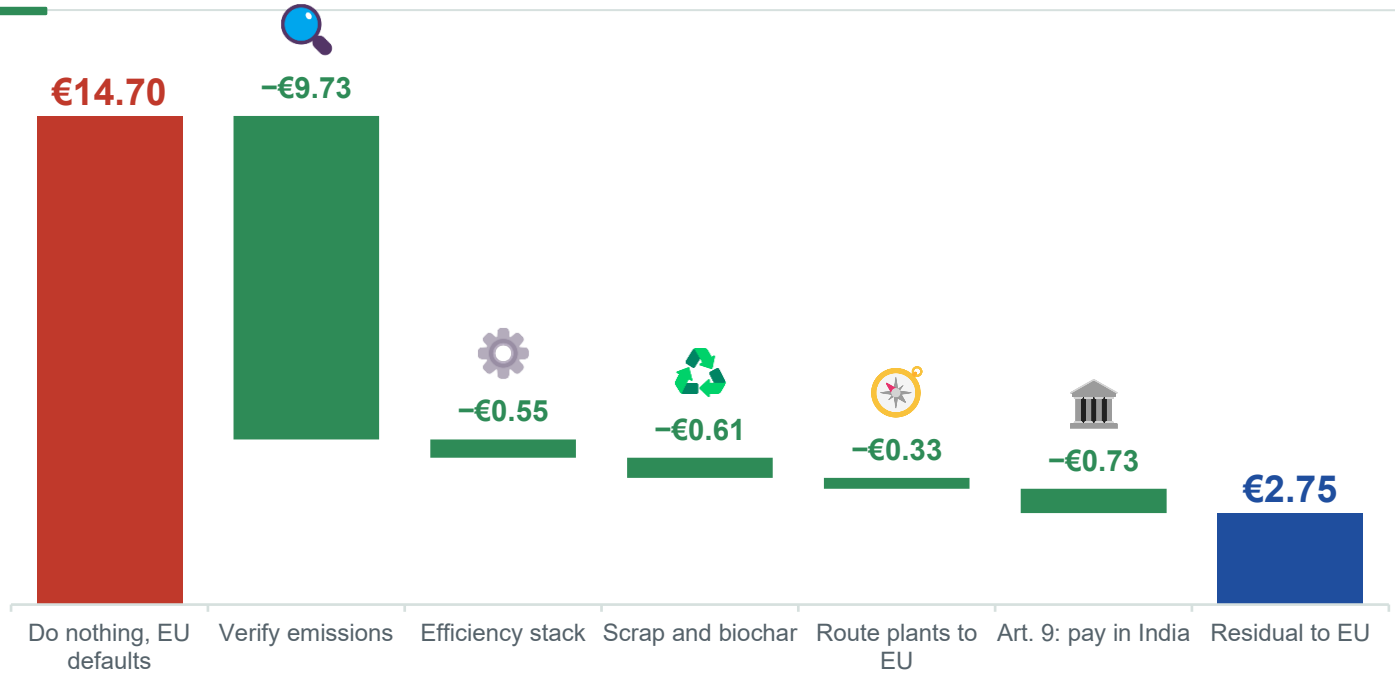
2026	2027	2028	2029	2030	2031	2032	2033	2034
97.5%	95%	90%	77.5%	51.5%	39%	26.5%	14%	0%

Verification removes €9.7bn of the €14.7bn; efficiency, scrap, routing and an Indian price take €2.2bn more, leaving €2.8bn

Sequential, so nothing is counted twice: each step is computed on what is left after the one before. Five exporters, 2026 to 2034, 100% pass through.

1 Value at stake, 2026 to 2034

€bn



66% of the default bill goes with verification alone. The residual €2.75bn is 55% of the verified bill of €4.97bn.

2 Route EU orders to the cleanest plant

JSW, 2030

Plant	CBAM €/t	Pro rata	Optimised
Dolvi	€133	0.48	1.30
Vijayanagar	€145	0.60	0.00
Sambalpur	€188	0.22	0.00

+€10m

a year, €34m to €44m of EU net premium

EU premium €100/t over other markets, 50% pass through. A greedy rank by net premium is optimal here (tested against brute force in the repo).

3 Verified bill by exporter, 2026 to 2034

€bn

JSW	€1.70bn (€5.19bn defaults)
AM/NS	€1.01bn (€2.90bn defaults)
Tata	€0.98bn (€3.56bn defaults)
Jindal	€0.90bn (€2.07bn defaults)
SAIL	€0.38bn (€0.99bn defaults)

Source: Moves: efficiency −0.2 t/t from 2027; scrap and biochar −0.25 t/t from 2028; routing by plant to the EU cap; explicit €20/t Indian price with 60% of tonnes qualifying under Art. 9. Plant intensities: CCTS baselines x 0.85 (grade D). cbam-india model (31 Oct 2026), ported and tied out in this deck's ledger.

All term moved

Verdict Do verification now: it is 81% of the value and needs no capex; the other four moves are worth doing but only together.

Next What government should do

Free allocation left

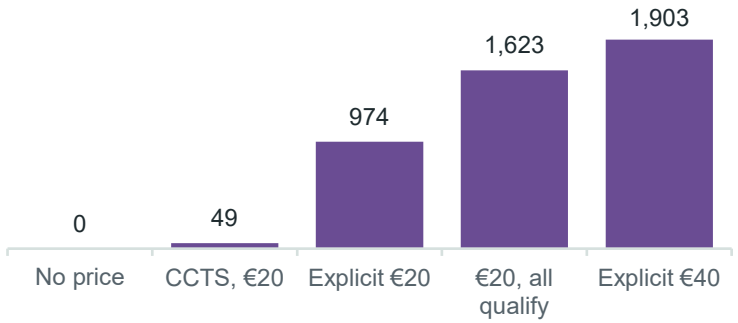
2026	2027	2028	2029	2030	2031	2032	2033	2034
97.5%	95%	90%	77.5%	51.5%	39%	26.5%	14%	0%

An explicit €20 a tonne Indian price keeps €974m at home; CCTS as designed keeps €49m, and $1.2bn of MSME goods is already in scope

Article 9 deducts a carbon price effectively paid in India. Steel scaled to 4.01 Mt of EU imports; 60% of tonnes carry accepted evidence.

1 CBAM kept in India, 2026 to 2034

€m, steel



Paid to the EU either way: €5.46bn with no price, €4.49bn with an explicit €20.

2 Carbon price ladder

€ per tCO2, 2026

- **€89.19** EU ETS allowance, spot 30 Oct 2026
- **€82.32** CBAM certificate, Q3 2026
- **€8.49** China national ETS
- **€6.03** Korea ETS, auction average
- **none** India CCTS, no price yet

Border gap: CBAM still collects €74 to 76 per chargeable tonne after crediting China or Korea.

3 Why CCTS keeps so little

t paid per t of steel

Explicit price on every embedded tonne

2.20 t

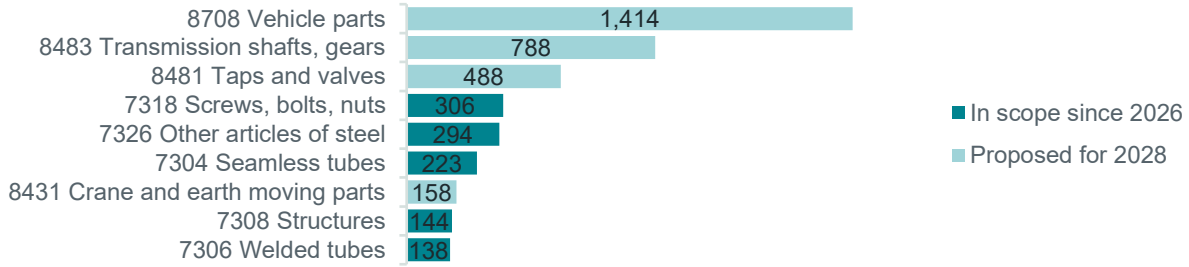
CCTS: only tonnes above a 5% target cut

0.11 t

CCTS: 490 entities with targets, 255 steel units in draft, certificates trading from November. The EU act requires a binding price, net of rebates.

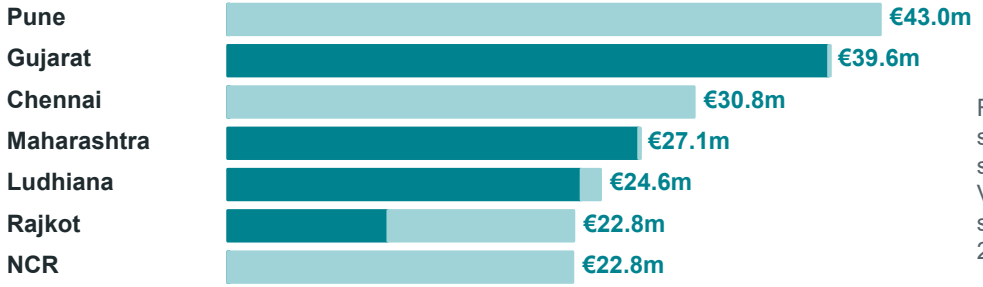
4 Downstream: $1.2bn in scope, $2.9bn proposed

EU imports from India 2024, $m



5 MSME clusters: CBAM on defaults, 2028

€m a year, proposal adopted



Fasteners, tubes, structures: in scope now. Vehicle parts, shafts, valves: 2028 if adopted.

Source: Repo policy.ts (article9, policyOptions, downstreamRows); price ladder: Trading Economics 30 Oct 2026, DG TAXUD, ICAP factsheets ($1.16 per €); downstream: UN Comtrade 2024, EU reporter 97, partner India 699 (grade B); COM(2025) 989 proposal; cluster split equal across listed clusters (grade D).

Art. 9
term moved

Verdict Price exported carbon explicitly and file the evidence with the EU; put MRV help desks in Ludhiana, Rajkot and Moradabad before 2028.

Next Roadmap

Free allocation left

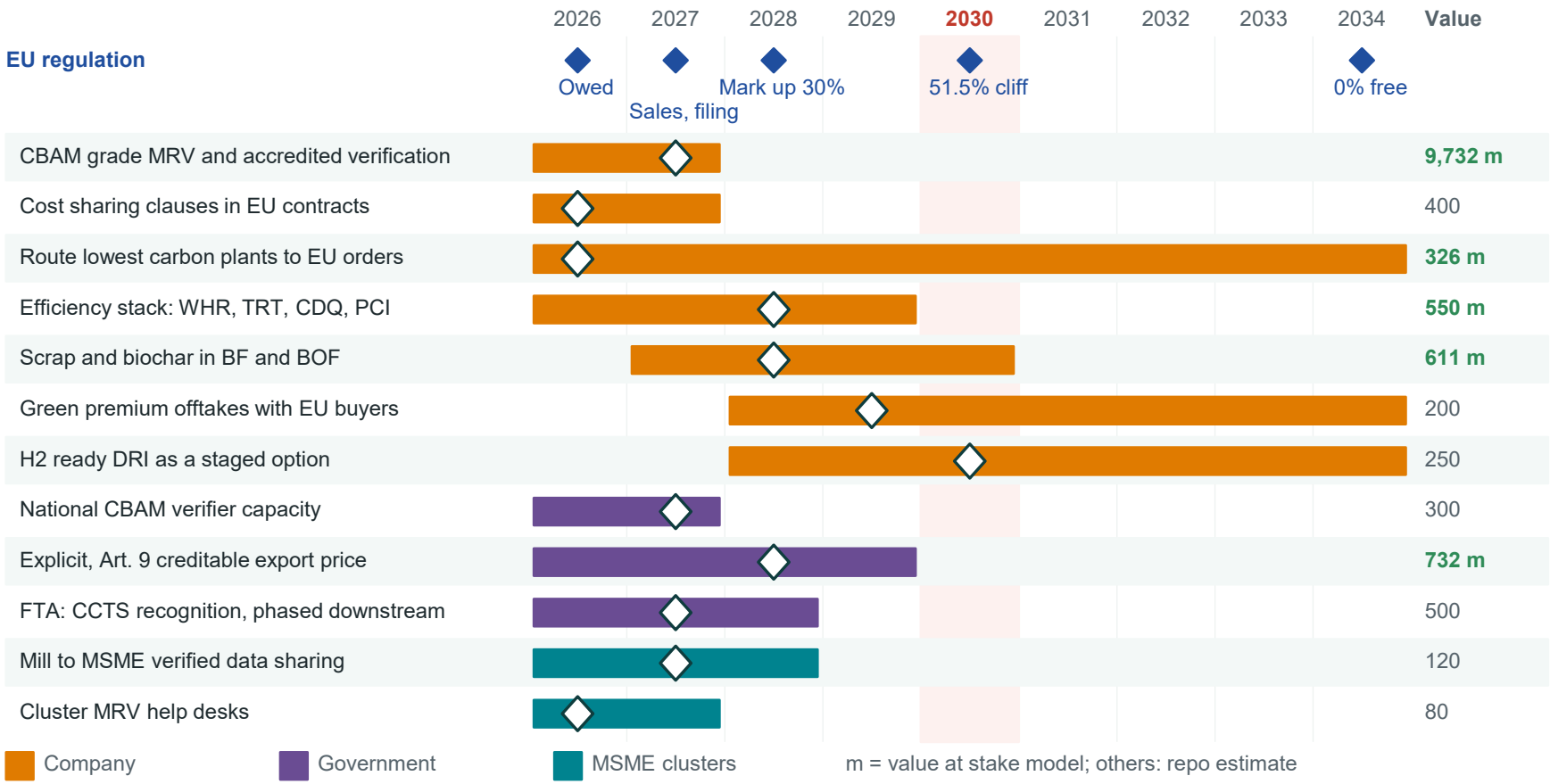
2026	2027	2028	2029	2030	2031	2032	2033	2034
97.5%	95%	90%	77.5%	51.5%	39%	26.5%	14%	0%

Twelve moves, front loaded before the 2030 cliff: verification, contract clauses and plant routing start in the next 90 days

Bars are active years, diamonds the milestone. Value is €m over 2026 to 2034: model computed where marked m, otherwise the repo's estimate.

1 Roadmap 2026 to 2034

owner colour; value €m



Company Government MSME clusters m = value at stake model; others: repo estimate

2 First 90 days

KPI

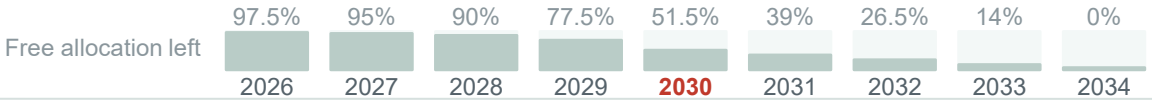
- Appoint a verifier; map installations to CN codes; file the first verified report
  KPI: % of EU tonnes verified: 0 to 100% by 2027
- Index the clause to the weekly certificate price; re sign the top 10 EU accounts
  KPI: exporter share of CBAM, from 50% base
- Put plant level CBAM in S&OP; route the next EU quarter and quota slots
  KPI: EU weighted t/t vs group average
- Fast track verifier accreditation with NABCB; open EU recognition talks
  KPI: accredited verifiers; days to verify
- Standard supplier data template; pilot in the Ludhiana fastener cluster
  KPI: % of MSME EU shipments with mill data

Source: Initiatives, timing and KPIs: repo storyline.ts (analyst proposal, not a model output); regulation track: repo timeline.ts plus 2026 events on page 7; values marked m are from the value at stake model on page 9.

All term moved

Verdict Nine of twelve moves start in 2026; anything that lands after 2029 meets the cliff at full price.

Next Appendix: method and limits



Method: one formula, nine factor years, eight producers; direct intensity moves the JSW bill most, then price and volume

certificates per t = max(0, E − F x B); CBAM € per t = certificates x P − Art. 9 credit; exporter hit = CBAM per t x pass through x EU volume.

1 Rule constants

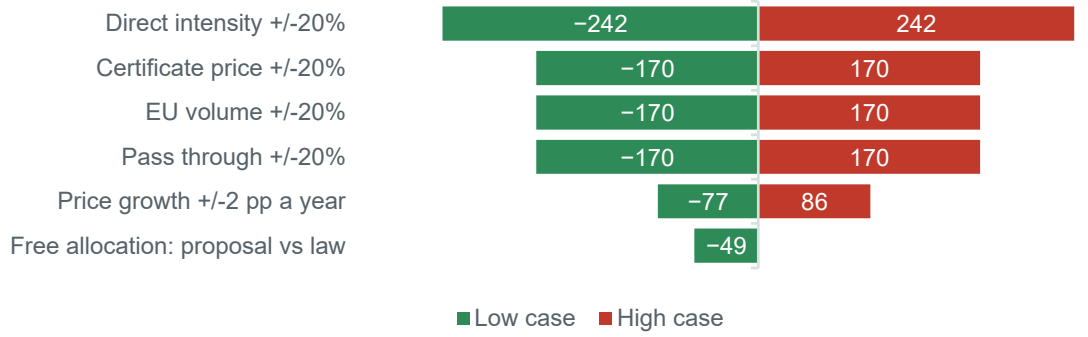
law unless marked

Year	2026	2027	2028	2029	2030	2031	2032	2033	2034
CBAM factor F	97.5%	95.0%	90.0%	77.5%	51.5%	39.0%	26.5%	14.0%	0.0%
Proposal (not law)	97.5%	95.0%	91.5%	80.9%	58.8%	48.2%	37.5%	26.9%	15.0%
Default mark up	10%	20%	30%	30%	30%	30%	30%	30%	30%
Price scenario €	82.3	84.8	87.3	90.0	92.7	95.4	98.3	101.2	104.3

Benchmarks (t per t): hot rolled BF-BOF 1.370, DRI-EAF 0.481, scrap EAF 0.072; primary aluminium 1.423. Defaults: India 7208 to 7210 4.28; aluminium 1.87; cement 1.48. Certificate 2026: Q1 €75.36, Q2 €75.28, Q3 €82.32.

2 Sensitivity of JSW's 2026 to 2034 exposure

€m change, base €851m at 50% pass through

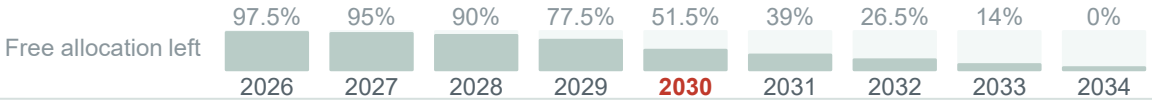


3 Producer inputs

grade in brackets

Producer	Production Mt (C)	EU volume Mt (D)	Direct t/t (D)	Benchmark t/t	Indirect t/t (D)	EBITDA €/t	2026 verified €/t	2026 defaults €/t	2030 verified €/t
JSW Steel	30.14	1.30	2.23	1.281	0.3	99	81	285	145
Tata Steel	23.48	0.90	2.01	1.370	0.3	150	56	278	121
AM/NS India	7.36	0.70	2.20	0.925	0.4	90	107	313	160
Jindal Steel	9.25	0.50	2.64	0.925	0.4	105	143	313	200
SAIL	19.43	0.25	2.55	1.370	0.3	68	100	278	171
Vedanta Aluminium	2.45	0.15	1.80	1.423	14.5	1,084	34	55	99
Hindalco	1.35	0.10	1.70	1.423	14.0	1,466	26	55	90
NALCO	0.46	0.06	1.75	1.423	14.5	1,100	30	55	94

Source: Reg. (EU) 2023/956; Directive 2003/87/EC Art. 10a; Impl. Regs (EU) 2025/2620 (benchmarks), 2025/2621 and 2026/1740 (defaults, mark ups); DG TAXUD certificate prices; inputs and grades: repo data.ts (A verified installation data, B regulation or official statistic, C company disclosure, D analyst estimate or derived, E Commission default); cbam-india model (31 Oct 2026), ported and tied out in this deck's ledger.



Every headline number ties to the repo within rounding; the weakest inputs are plant intensities and EU volumes, both grade D

The ledger re implements the repo engine in Python (model_cbam.py) and checks it against the repo's published numbers.

1 Tie out: repo against deck ledger

13 of 13 within rounding

Number	Repo	Ledger	
do nothing € bn	14.7	14.70	ok
residual € bn	2.8	2.75	ok
Tata default €/t 2026	278	277.60	ok
Tata verified €/t 2026	56	55.50	ok
verification € m a year	739	739.13	ok
verification share %	66	66.18	ok
India verified 2030 €/t	129	129.20	ok
Korea verified 2030 €/t	120	119.93	ok
India default penalty 2030	321	320.95	ok
explicit € bn kept	1	0.97	ok
CCTS € m kept	49	48.68	ok
MSME in scope $ bn	1.2	1.19	ok
downstream proposed $ bn	2.9	2.89	ok

2 What the data does not support

read before quoting

⚠️ Plant intensities are CCTS baselines x 0.85 (grade D), not CBAM verified data. One verified report per installation replaces them.

⚠️ EU volumes per producer are estimates (grade D), and are held flat to 2034. H1 FY27 shipments to the EU are down 19.8% and the quota now caps them.

⚠️ Pass through is a slider (50% base). The legal payer is the EU importer; the exporter's share is a commercial outcome.

📜 The Article 9 act is not yet adopted; the €20 explicit price and 60% qualifying share are design choices, not policy.

🔍 Counterview: Sandbag (Aug 2026) finds the impact on Indian exports could be far smaller than feared, as some plants are clean enough to gain from higher EU prices.

📊 Delivered costs, FOB and freight are illustrative 2026 levels; the crossover prices move one for one with them.

❌ Not modelled: the 50% out of quota duty, EU price response, weekly certificate prices from 2027, product mix beyond CN 7208.

Source: cbam-india README and faq.ts (31 Oct 2026); deck ledger model_cbam.py (9 Nov 2026); Sandbag, The EU CBAM's real impact on India (Aug 2026); Nature Climate Change, Vriz et al. (Jun 2026); LSE Grantham (2026); Carbon Pulse (Oct 2026); EUR-Lex.